COMP 3311: Database Management Systems

Lecture 11 Exercises
Structured Query Language (SQL): DDL and DML

Author(<u>authorId</u>, firstName, lastName)	Book(<u>bookId</u>, title, subject, quantityInStock, price, *authorId*)
BookOrder(<u>orderId</u>, *customerId*, orderYear)	Customer(<u>customerId</u>, firstName, lastName)
OrderDetails(<u>*orderId, bookId*</u>, quantity)	

1. **<u>Download</u>** the zipped folder **L11ExerciseFiles.zip** from Canvas and unzip it. The folder contains four SQL script files **bookstoredb.sql**, **bookstoreOrderData.sql**, **L11Exercises1-4.sql** and **L11Exercises5-7.sql**.
2. **<u>Execute</u>** the **bookstoredb.sql** script file in **Oracle SQL Developer** to create the database against which you can run your exercise queries.
3. **<u>Modify</u>** the **L11Exercises5-7.sql** script file by constructing your SQL queries and statements in the specified locations in the script file.
4. **<u>Test</u>** your SQL queries and statements to ensure that they do not contain any errors.

How to test Exercise 1

After you have executed both your alter table statement and BestCustomers view in **Oracle SQL Developer**

1. **<u>run the query</u>** that retrieves all the tuples in the Customer relation.
   - The value of the numOrders attribute should be 0 for all customers.
2. **<u>run the query</u>** that retrieves all the tuples in the BestCustomers view.
   - The BestCustomers view should contain only the customer Harry Lee.
3. **<u>insert</u>** the tuple (7, 'Rusty', 'Zhang', 0) into the BestCustomers view.
4. **<u>run again</u>** the queries that retrieve all the tuples in the Customer relation and in the BestCustomers view.
   - The tuple for Rusty Zhang <u>should</u> appear in the Customers relation with a null hkid and <u>should not</u> appear in the BestCustomers view.

How to test Exercise 2

After you have executed your IncrementOrders trigger in **Oracle SQL Developer**

a. **<u>run</u>** the **bookstoreOrderData.sql** script file.

b. **<u>run again</u>** the query that retrieves all the tuples in the Customer relation.
   - The number of orders for customer Ron Lee should be 5.

c. **<u>insert</u>** the tuple (26, 3, '2022') into the BookOrder relation.

d. **<u>run again</u>** the query that retrieves all the tuples in the BestCustomers relation.
   - Customer Ron Lee should now appear in the query result and the number of orders for him should be 6.

This is an <u>individual</u> exercise. EACH STUDENT MUST SUBMIT THEIR OWN SQL SCRIPT FILE.
A genuine effort must be made to complete <u>all</u> exercises to obtain full marks.
USE ONLY SQL CONSTRUCTS DISCUSSED IN THE COURSE.

What To Submit

Your modified **L11Exercises5-7.sql** script file containing your SQL queries and statements.

You must submit this completed script file. No other type of file submission is acceptable.

How To Submit

By 6:00 p.m. tomorrow upload your modified **L11Exercises5-7.sql** script file to Canvas.

Note: You will only be able to submit a file with extension sql.

> **TEST YOUR SCRIPT FILE – USE ONLY SQL CONSTRUCTS DISCUSSED IN THE COURSE**
>
> The SQL queries in your submission will be tested by executing them against the **bookstoredb** database. If any of the queries raises an SQL error or if you use SQL constructs not discussed in the course, then you will receive at most 0.5 marks for this exercise. Therefore, test your SQL queries against the **bookstoredb** database before you submit them and use only SQL constructs discussed in the course.